Session 8: Introduction to PL/SQLTrainer Coverage:PL/SQL block structure (DECLARE, BEGIN, EXCEPTION, END).Variables and cursors.Control structures (IF, LOOP).Demo: Simple stored procedure to calculate user's total spend.Pending

Tasks

1) **Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.**

Query:

Run SQL query/queries on database my database:

```
CREATE PROCEDURE show_user()
BEGIN
    DECLARE user_name VARCHAR(50);
    DECLARE user_age INT;

    SET user_name = 'Abhi';
    SET user_age = 20;

    SELECT user_name AS 'User Name',
           user_age AS 'Age';
END //

DELIMITER ;

CALL show_user();
```

Output:

Extra options

User Name	Age
Abhi	20

2) **Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply'.**

Query:

Run SQL query/queries on database my database:

```
DELIMITER //

CREATE PROCEDURE check_delivery()
BEGIN
    DECLARE order_amount DECIMAL(10,2);

    SET order_amount = 600;

    IF order_amount > 500 THEN
        SELECT 'Eligible for free delivery' AS Message;
    ELSE
        SELECT 'Delivery charges apply' AS Message;
    END IF;
END //

DELIMITER ;
```

Output:

Extra options

Message
Eligible for free delivery

3) **Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using DBMS_OUTPUT.PUT_LINE.

Hint: Use a counter variable and EXIT WHEN condition inside the loop.**

Query:

Run SQL query/queries on database my database:

```
CREATE PROCEDURE print_numbers()
BEGIN
    DECLARE counter INT DEFAULT 1;

    simple_loop: LOOP
        SELECT counter;

        SET counter = counter + 1;

        IF counter > 5 THEN
            LEAVE simple_loop;
        END IF;
    END LOOP simple_loop;
END //

DELIMITER ;
```

Output:

Routines

Check all | Export | Drop

	Name	Type	Returns				
☐	check_delivery	PROCEDURE		Edit	Execute	Export	Drop
☐	print_numbers	PROCEDURE		Edit	Execute	Export	Drop
☐	show_user	PROCEDURE		Edit	Execute	Export	Drop

4) **Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE.

Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.**

QUERY:

Run SQL query/queries on database my database:

```sql
CREATE TABLE PRODUCTS (
    PRODUCT_ID INT PRIMARY KEY AUTO_INCREMENT,
    PRODUCT_NAME VARCHAR(100)
);

INSERT INTO PRODUCTS (PRODUCT_NAME) VALUES
('Laptop'),
('Mobile Phone'),
('Keyboard'),
('Mouse'),
('Headphones');
```

Run SQL query/queries on database my database:

```sql
CREATE PROCEDURE show_products()
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE p_name VARCHAR(100);

    DECLARE product_cursor CURSOR FOR
        SELECT PRODUCT_NAME FROM PRODUCTS;

    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    OPEN product_cursor;

    product_loop: LOOP
        FETCH product_cursor INTO p_name;

```

```sql
        END IF;

        SELECT p_name AS Product_Name;
    END LOOP;

    CLOSE product_cursor;
END //

DELIMITER ;
```

OUTPUT:

Extra options

				PRODUCT_ID	PRODUCT_NAME
☐	Edit	Copy	Delete	1	Laptop
☐	Edit	Copy	Delete	2	Mobile Phone
☐	Edit	Copy	Delete	3	Keyboard
☐	Edit	Copy	Delete	4	Mouse
☐	Edit	Copy	Delete	5	Headphones

Routines

Check all | Export | Drop

	Name	Type	Returns				
☐	check_delivery	PROCEDURE		Edit	Execute	Export	Drop
☐	print_numbers	PROCEDURE		Edit	Execute	Export	Drop
☐	show_products	PROCEDURE		Edit	Execute	Export	Drop
☐	show_user	PROCEDURE		Edit	Execute	Export	Drop

5) **Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.**

QUERY:

```

CREATE PROCEDURE divide_numbers()
BEGIN
    DECLARE num1 DECIMAL(10,2) DEFAULT 10;
    DECLARE num2 DECIMAL(10,2) DEFAULT 0;
    DECLARE result DECIMAL(10,2);

    DECLARE CONTINUE HANDLER FOR SQLSTATE '22012'
    BEGIN
        SELECT 'Cannot divide by zero' AS Message;
    END;

    SET result = num1 / num2;

    SELECT result AS Result;
```

OUTPUT: